

Proposal for Car Accident Risk Prediction Solution

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1. Introduction

This proposal presents our technical approach to developing an AI-driven Car Accident Risk Prediction Tool from the ground up. Unlike previous iterations, our solution will be built entirely from scratch, including model development and data collection, eliminating reliance on older datasets and models (though we will consult and review past data for insights). Our goal is to provide a robust, threshold-driven risk alert mechanism that seamlessly integrates with your mobile application via a dedicated API endpoint.

2. Objectives

Our proposed solution aims to:

- Construct a brand-new dataset tailored for precise car accident risk prediction.
- Develop a high-fidelity rule-based classifier with an ML fallback if precision drops below a 70% threshold.
- Establish API endpoints to enable seamless integration and testing with your frontend mobile application.
- Ensure rapid implementation with a two-week timeline and a cost-efficient budget of \$500.
- Utilize evaluation metrics to rigorously test the model's accuracy and effectiveness.
- Build an API capable of simulating the alert system for enhanced real-time risk assessment.

3. Proposed Solution

We will deploy a rule-based classifier as the primary mechanism, leveraging heuristic thresholds (e.g., accident frequency, weather conditions, and location-based risk factors). The classifier will generate alerts only when risk levels exceed predefined bounds. If model performance falls below the 70% precision benchmark, we will implement an ML-based solution using advanced algorithms (e.g., ensemble methods, gradient boosting). The solution will feature:

- **End-to-End Data Collection:** Sourcing and curating structured accident data from trusted repositories, including government sources and third-party APIs.
- **Rule-Based Classifier with ML Backup:** Ensuring accuracy while maintaining computational efficiency.

- **API Endpoint for Mobile Integration:** Providing a user-friendly interface for real-time risk assessment.
- **Consultation on Past Data:** While eliminating reliance on outdated models, we will review prior datasets for insights.
- **Evaluation Metrics:** The rule-based classifier will be validated using statistical measures such as precision, recall, F1-score, and accuracy by comparing predictions against historical accident data. We will conduct threshold tuning to optimize risk classification and assess false positive/negative rates. Additionally, a test suite with simulated scenarios will be implemented to validate system performance before deployment.
- **Simulated Alert System API:** Developing an API that can emulate real-time accident risk alerts.

4. Scope of Work: Milestone 1

Our streamlined approach will focus on three key phases:

Phase	Tasks	Timeline	Cost (USD)	Resources
Data Collection	Gather and structure accident datasets	Week 1	\$200	Data Engineers
Model Development	Build and test rule-based/ML classifier	Week 1-2	\$250	Data Scientists
API Development	Deploy API for mobile integration	Week 2	\$100	Solution Architects
Total		2 weeks	\$550	

5. Deliverables

- **Dataset Report:** A newly built dataset for accident risk prediction.
- **Model Documentation:** Full specifications for the implemented rule-based or ML classifier.
- **API Endpoint:** A functional API to facilitate seamless mobile app integration.
- **Technical Roadmap:** A comprehensive guide outlining future system enhancements.
- **Evaluation Metrics Report:** Detailed analysis of model performance and validation.
- **Simulated Alert System API:** A dedicated API to test real-time alert functionality.

6. Key Features

- **Threshold-Driven Alerts:** Engages only when accident risk surpasses calibrated thresholds.
- **New Data Infrastructure:** Designed from scratch for optimal accuracy and reliability.
- **Mobile-Friendly API:** Ensuring smooth frontend integration and testing.
- **Consultation on Past Data:** Insights derived from older datasets without full reliance.

- **Evaluation Metrics:** Ensuring the model meets rigorous accuracy standards.
- **API for Simulated Alerts:** Enabling real-time testing and refinement of risk assessment mechanisms.